Orbital Station Relativistic Momentum, Plasma Sheath Electrodynamics, and Hamiltonian Orbital Mechanics

Advanced Undergraduate Physics & Aerospace Systems Seminar

Prepared by CERN

August 12, 2026

Problem Statement

A next-generation low-altitude equatorial orbital station of mass M_0 deploys an autonomous tethered electrodynamic propulsion module through a non-uniform planetary magnetosphere $\vec{B}(\vec{r})$ while encountering a non-isothermal plasma ionosphere.

The background atmospheric plasma density and ion temperature follow double-exponential profiles:

$$n_e(z) = n_0 \exp\left(-\frac{z - z_0}{H_e}\right), \quad T_i(z) = T_0 \left[1 + \eta \exp\left(-\frac{z}{H_T}\right)\right] \tag{1}$$

where n_0 is base plasma density, H_e is plasma scale height, T_0 is ion baseline temperature, and H_T is thermal scale height.

The module executes precise orbital plane transitions using electrodynamic tether forces combined with high-impulse magnetoplasmadynamic (MPD) thrust. To maintain continuous orientation control during orbital maneuvers, the onboard flight computer evaluates 3D Hamiltonian mechanics, relativistic four-momentum conservation, and plasma sheath Debye-length dynamics.

Part A: Relativistic Kinematics, Four-Vectors, and Angular Momentum Tensors

Question 1: Minkowski Four-Momentum Mechanics & Mass Invariance

Let $p^\mu = (E/c, \vec{p}) \in \mathbb{R}^{1,3}$ be the relativistic four-momentum of a high-energy particle emitted from an MPD plasma thruster in the station's rest frame, where $E = \gamma mc^2$ and $\vec{p} = \gamma m\vec{v}$ with Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$.

(a) Prove using Minkowski metric tensor $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$ that the inner product $\langle p, p\rangle = \eta_{\mu\nu}p^\mu p^\nu$ is a Lorentz scalar invariant strictly equal to $-m^2c^2$.

(b) Derive the explicitly symmetric-skew decomposition of the relativistic angular momentum tensor $L^{\mu\nu} = x^\mu p^\nu - x^\nu p^\mu$ and prove that its spatial-spatial components L^{ij} directly reduce to the classical 3D orbital angular momentum vector components $\vec{L} = \vec{r} \times \vec{p}$.

Question 2: Relativistic Doppler Shift Vector Geometry

Let an optical tracking station on the planetary surface transmit a laser guidance signal with four-wavevector $k^\mu = (\omega/c, \vec{k})$ to the orbital station moving at velocity vector $\vec{v}$ relative to the ground transmitter.

(a) Construct the Lorentz transformation matrix $\Lambda^\mu{}_\nu(\vec{v})$ for an arbitrary spatial direction $\hat{\beta} = \vec{v}/c$.

(b) Evaluate the scalar product $k_\mu u^\mu$ (where u^μ is the station four-velocity) to derive the exact 3D relativistic vector Doppler shift equation for observed frequency ω' as a function of emission angle $\theta = \arccos(\hat{k} \cdot \hat{\beta})$.

Part B: Plasma Sheath Electrodynamics & Line-Integral Power Losses

Question 3: Electrodynamic Tether Current Induction & Sheath Jacobians

An electrodynamic tether of length L_t oriented along unit vector $\hat{r}_t$ cuts through the planetary magnetic field $\vec{B}(\vec{r})$ at velocity $\vec{v}_s$. The motional electromotive force generates an induced voltage $V_{\text{emf}} = \int_0^{L_t} (\vec{v}_s \times \vec{B}) \cdot d\vec{l}$.

(a) Compute the 3×3 Jacobian matrix mapping linear station position variations $\vec{r} = [x, y, z]^T$ to magnetic induction variations $\vec{B}(\vec{r}) = \frac{\mu_0 M_m}{4\pi r^5}[3xz, 3yz, 2z^2 - x^2 - y^2]^T$ (dipole field approximation).

(b) Evaluate $J_{\vec{B}} = \frac{\partial \vec{B}}{\partial \vec{r}}$ explicitly at the equatorial orbit locus $\vec{r}_0 = [R_e, 0, 0]^T$.

Question 4: Debye Sheath Line-Integral Energy Dissipation

The thermal electron collection current per unit area along the conducting tether surface obeys the Langmuir-Mott-Smith probe equation in a magnetized plasma:

$$j_e(z) = en_e(z)\sqrt{\frac{k_BT_e}{2\pi m_e}}\left(1 + \frac{eV(z)}{k_BT_e}\right)^{1/2} \tag{2}$$

(a) Consider an orbital decay arc parameterized by time $t \in [0, T_f]$ where altitude decreases linearly $z(t) = z_0 - v_z t$ and tether potential varies quadratically $V(t) = V_0(1 - \alpha t)^2$. Evaluate the total integrated electrical energy generated $E_{\text{total}} = A_{\text{tether}} \int_0^{T_f} j_e(t)V(t)\, dt$ in closed definite integral form.

Part C: Hamiltonian Dynamics, Canonical Transformations, and State Stability

Question 5: Planar Orbital Hamiltonian & Linearized Perturbation State-Space

The 2D planar orbital motion of the station under planetary gravity $\mu = GM_p$ and continuous low-thrust acceleration $\vec{a}_t = [a_r, a_\theta]^T$ is expressed in polar canonical coordinates $(r, \theta, p_r, p_\theta)$:

$$H(r, \theta, p_r, p_\theta) = \frac{p_r^2}{2M_0} + \frac{p_\theta^2}{2M_0 r^2} - \frac{GM_p M_0}{r} \tag{3}$$

(a) Derive Hamilton's canonical equations of motion $\dot{r} = \frac{\partial H}{\partial p_r}$, $\dot{p}_r = -\frac{\partial H}{\partial r}$, $\dot{\theta} = \frac{\partial H}{\partial p_\theta}$, $\dot{p}_\theta = -\frac{\partial H}{\partial \theta}$ and prove that angular momentum $p_\theta = M_0 r^2 \dot{\theta}$ is a strict constant of motion when external thrust $a_\theta = 0$.

(b) Linearize the radial Hamiltonian state-space system around a circular equilibrium orbit $r_0 = \left(\frac{p_{\theta 0}^2}{GM_p M_0^2}\right)$ with small perturbations $x = [\Delta r, \Delta p_r]^T \in \mathbb{R}^2$ to construct the system matrix $A \in \mathbb{R}^{2\times 2}$ such that $\dot{x} = Ax$.

(c) Compute the eigenvalues $\lambda_{1,2}$ of system matrix A, determine the natural orbital resonance frequency ω_0, and apply full state-feedback control $u_c = -Kx$ with $K = [k_1, k_2]$ to place closed-loop poles at prescribed stable real values $\{\mu_1, \mu_2\} \subset \mathbb{R}^-$.

Part D: Strategic Mindmap & Mental Framework

Meta-Strategy: Conceptual Deconstruction

To solve high-dimensional space systems problems spanning special relativity, plasma physics, vector calculus, and analytical dynamics, structure your analysis into three sequential passes:

1. **Relativistic & Kinematic Pass (Part A):** Establish invariant scalar quantities ($p_\mu p^\mu$), tensor symmetries ($L^{\mu\nu}$), and frame transformation matrices ($\Lambda^\mu{}_\nu$).
2. **Field & Electrodynamic Pass (Part B):** Compute vector partial derivative matrices (Jacobians $J_{\vec{B}}$) and integrate non-linear kinetic fields along descent curves.
3. **Analytical Mechanics Pass (Part C):** Construct the Hamiltonian $H(q, p)$, compute canonical field derivatives, linearize around circular equilibria, and synthesize closed-loop state feedback.

1. Mindmap for Part A: Relativistic Kinematics & Vectors

Problem 1: Four-Momentum Invariance & Angular Momentum Tensors

- **Mental Picture:** A high-velocity plasma particle moving through Minkowski spacetime. The four-momentum vector length is invariant across all inertial frames.
- **Mathematical Tool:** Metric tensor contraction $\eta_{\mu\nu}$ and skew-symmetric tensor decomposition.
- **Approach Strategy:**
  1. Write $p^\mu = (\gamma mc, \gamma m\vec{v})$. Contract with $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ to get $-\gamma^2 m^2 c^2 + \gamma^2 m^2 v^2$. Factor $\gamma^2(v^2 - c^2) = -c^2$.
  2. Expand $L^{ij} = x^i p^j - x^j p^i$ for spatial indices $i, j \in \{1, 2, 3\}$ and match terms directly to $\vec{r} \times \vec{p}$.

Problem 2: Relativistic Doppler Wavevectors

- **Mental Picture:** Laser light wave-fronts intercepted by a high-speed orbital platform.
- **Mathematical Tool:** Lorentz boost matrix transformations and scalar invariants.
- **Approach Strategy:**
  1. Write station four-velocity $u^\mu = (\gamma c, \gamma\vec{v})$.
  2. Evaluate scalar invariant $k_\mu u^\mu = -\frac{\omega}{c}(\gamma c) + \vec{k} \cdot (\gamma\vec{v})$.
  3. Set equal to frame-invariant frequency $-\omega'$ to derive $\omega' = \gamma\omega(1 - \beta\cos\theta)$.

2. Mindmap for Part B: Plasma Electrodynamics & Integrals

Problem 3: Magnetic Dipole Jacobian Field Mapping

- **Mental Picture:** A satellite moving through a curved planetary magnetic dipole field.
- **Mathematical Tool:** Multivariable vector partial differentiation $\frac{\partial B_i}{\partial x_j}$.
- **Approach Strategy:**
  1. Compute partial derivatives of B_x, B_y, B_z with respect to x, y, z using quotient rules.
  2. Evaluate all entries at equatorial position $[R_e, 0, 0]^T$.

Problem 4: Non-Linear Plasma Sheath Energy Integration

- **Mental Picture:** A tether collecting plasma electrons while descending through atmospheric layers.
- **Mathematical Tool:** Definite path integration with exponential and polynomial terms.
- **Approach Strategy:**
  1. Substitute $z(t)$ into $n_e(z)$ to form exponential factor $\exp\left(\frac{v_z t}{H_e}\right)$.
  2. Expand potential term $V(t)^{3/2} = V_0^{3/2}(1 - \alpha t)^3$.
  3. Integrate product of exponential and polynomial functions over time $[0, T_f]$.

3. Mindmap for Part C: Hamiltonian Mechanics & State Control

Problem 5: Canonical Orbital Mechanics & Pole Assignment

- **Mental Picture:** Planetary orbit analyzed in phase space (r, p_r), perturbed from circular equilibrium, and stabilized using active control thrusters.
- **Mathematical Tool:** Hamilton's canonical equations, Jacobian linearization, and characteristic polynomials.
- **Approach Strategy:**
  1. Compute $\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_r}{M_0}$ and $\dot{p}_r = -\frac{\partial H}{\partial r} = \frac{p_\theta^2}{M_0 r^3} - \frac{G M_p M_0}{r^2}$.
  2. Find equilibrium r_0 where $\dot{p}_r = 0$.
  3. Take partial derivatives of $[\dot{r}, \dot{p}_r]^T$ with respect to $[r, p_r]$ to construct A.
  4. Compute $\det(\lambda I - (A - BK)) = 0$ and match coefficients with desired polynomial $(\lambda - \mu_1)(\lambda - \mu_2) = 0$.

Part E: Complete Step-by-Step Mathematical Solutions

Problem 1: Relativistic Minkowski Four-Momentum & Tensor Proofs

(a) Invariance of Four-Momentum Norm $\langle p, p \rangle = -m^2c^2$

The relativistic four-momentum vector is $p^\mu = \begin{bmatrix} \gamma mc & \gamma mv_x & \gamma mv_y & \gamma mv_z \end{bmatrix}^T$. The Minkowski metric tensor is $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$.

Evaluating the scalar inner product $\eta_{\mu\nu} p^\mu p^\nu$:

$$\begin{aligned}
\langle p, p \rangle &= \eta_{00}(p^0)^2 + \eta_{11}(p^1)^2 + \eta_{22}(p^2)^2 + \eta_{33}(p^3)^2 && (4)\\
&= -(\gamma mc)^2 + (\gamma mv_x)^2 + (\gamma mv_y)^2 + (\gamma mv_z)^2 && (5)\\
&= -\gamma^2 m^2 c^2 + \gamma^2 m^2 (v_x^2 + v_y^2 + v_z^2) && (6)\\
&= -\gamma^2 m^2 c^2 + \gamma^2 m^2 v^2 && (7)\\
&= -\gamma^2 m^2 c^2 \left(1 - \frac{v^2}{c^2}\right) && (8)
\end{aligned}$$

Substituting the definition of the Lorentz factor $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \implies \gamma^2 \left(1 - \frac{v^2}{c^2}\right) = 1$:

$$\langle p, p \rangle = -m^2 c^2 \tag{9}$$

Since m (rest mass) and c (speed of light) are Lorentz frame-independent constants, $\langle p, p \rangle$ is a scalar invariant.

(b) Relativistic Angular Momentum Tensor $L^{\mu\nu}$ Spatial Components

The tensor is defined as $L^{\mu\nu} = x^\mu p^\nu - x^\nu p^\mu$. For spatial components $i, j \in \{1, 2, 3\}$ corresponding to coordinates (x, y, z):

$$L^{ij} = x^i p^j - x^j p^i \tag{10}$$

Evaluating specific spatial components:

$$\begin{aligned}
L^{12} &= x^1 p^2 - x^2 p^1 = xp_y - yp_x = L_z && (11)\\
L^{23} &= x^2 p^3 - x^3 p^2 = yp_z - zp_y = L_x && (12)\\
L^{31} &= x^3 p^1 - x^1 p^3 = zp_x - xp_z = L_y && (13)
\end{aligned}$$

Expressing in vector form:

$$\begin{bmatrix} L^{23} \\ L^{31} \\ L^{12} \end{bmatrix} = \begin{bmatrix} yp_z - zp_y \\ zp_x - xp_z \\ xp_y - yp_x \end{bmatrix} = \vec{r} \times \vec{p} = \vec{L} \tag{14}$$

Thus, the spatial block of $L^{\mu\nu}$ represents the 3D orbital angular momentum vector.

Problem 2: Vector Relativistic Doppler Shift Derivation

(a) General Lorentz Boost Matrix Construction

For velocity vector $\vec{v} = c\vec{\beta}$, let $\beta = \|\vec{\beta}\|_2$ and $\gamma = (1 - \beta^2)^{-1/2}$. The Lorentz boost transformation matrix $\Lambda^\mu{}_\nu$ is:

$$\Lambda^\mu{}_\nu = \begin{bmatrix} \gamma & -\gamma\beta_x & -\gamma\beta_y & -\gamma\beta_z \\ -\gamma\beta_x & 1 + (\gamma - 1)\frac{\beta_x^2}{\beta^2} & (\gamma - 1)\frac{\beta_x\beta_y}{\beta^2} & (\gamma - 1)\frac{\beta_x\beta_z}{\beta^2} \\ -\gamma\beta_y & (\gamma - 1)\frac{\beta_x\beta_y}{\beta^2} & 1 + (\gamma - 1)\frac{\beta_y^2}{\beta^2} & (\gamma - 1)\frac{\beta_y\beta_z}{\beta^2} \\ -\gamma\beta_z & (\gamma - 1)\frac{\beta_x\beta_z}{\beta^2} & (\gamma - 1)\frac{\beta_y\beta_z}{\beta^2} & 1 + (\gamma - 1)\frac{\beta_z^2}{\beta^2} \end{bmatrix} \tag{15}$$

(b) Scalar Invariant Evaluation of Observed Doppler Frequency

The four-wavevector is $k^\mu = \left(\frac{\omega}{c}, \vec{k}\right)$ where $\|\vec{k}\|_2 = \frac{\omega}{c}\hat{k}$. The station four-velocity is $u^\mu = (\gamma c, \gamma\vec{v})$.

Evaluating the invariant scalar product $g_{\mu\nu}k^\mu u^\nu$ in the transmitter frame:

$$\begin{aligned}
k_\mu u^\mu &= \eta_{\mu\nu}k^\mu u^\nu = -k^0u^0 + \vec{k}\cdot\vec{u} && (16)\\
&= -\left(\frac{\omega}{c}\right)(\gamma c) + \left(\frac{\omega}{c}\hat{k}\right)\cdot(\gamma\vec{v}) && (17)\\
&= -\gamma\omega + \gamma\frac{\omega}{c}\|\vec{v}\|_2\cos\theta && (18)\\
&= -\gamma\omega(1-\beta\cos\theta) && (19)
\end{aligned}$$

In the rest frame of the moving orbital station receiver, $u'^\mu = (c, \vec{0})$ and $k'^\mu = \left(\frac{\omega'}{c}, \vec{k}'\right)$. Evaluating the scalar product in the receiver frame:

$$k'_\mu u'^\mu = -\left(\frac{\omega'}{c}\right)(c) + \vec{k}'\cdot\vec{0} = -\omega' \tag{20}$$

Equating frame invariants ($k_\mu u^\mu = k'_\mu u'^\mu$):

$$-\omega' = -\gamma\omega(1-\beta\cos\theta) \implies \omega' = \gamma\omega(1-\beta\cos\theta) \tag{21}$$

Expressing explicitly in terms of velocity ratio $\beta = v/c$:

$$\omega' = \frac{1-\frac{v}{c}\cos\theta}{\sqrt{1-\frac{v^2}{c^2}}}\,\omega \tag{22}$$

Problem 3: Magnetic Dipole Field Jacobian Matrix

The magnetic field vector of a dipole centered at the origin is:

$$\vec{B}(x,y,z) = \frac{\mu_0 M_m}{4\pi r^5}\begin{bmatrix} 3xz \\ 3yz \\ 2z^2 - x^2 - y^2 \end{bmatrix} \tag{23}$$

where $r = (x^2+y^2+z^2)^{1/2}$. Let $K_0 = \frac{\mu_0 M_m}{4\pi}$.

We compute the 3×3 Jacobian matrix $J_{\vec{B}} = \begin{bmatrix} \frac{\partial B_x}{\partial x} & \frac{\partial B_x}{\partial y} & \frac{\partial B_x}{\partial z} \\ \frac{\partial B_y}{\partial x} & \frac{\partial B_y}{\partial y} & \frac{\partial B_y}{\partial z} \\ \frac{\partial B_z}{\partial x} & \frac{\partial B_z}{\partial y} & \frac{\partial B_z}{\partial z} \end{bmatrix}$.

1. Partial Derivatives at Equatorial Position $\vec{r}_0 = [R_e, 0, 0]^T$

At $\vec{r}_0 = [R_e, 0, 0]^T$: $x = R_e$, $y = 0$, $z = 0$, and $r = R_e$.

Evaluating each entry:

$$\frac{\partial B_x}{\partial x} = K_0 \left[\frac{3z}{r^5} - \frac{5(3xz)x}{r^7} \right]_{\vec{r}_0} = 0 \tag{24}$$

$$\frac{\partial B_x}{\partial y} = K_0 \left[0 - \frac{5(3xz)y}{r^7} \right]_{\vec{r}_0} = 0 \tag{25}$$

$$\frac{\partial B_x}{\partial z} = K_0 \left[\frac{3x}{r^5} - \frac{5(3xz)z}{r^7} \right]_{\vec{r}_0} = \frac{3K_0 R_e}{R_e^5} = \frac{3K_0}{R_e^4} \tag{26}$$

$$\frac{\partial B_y}{\partial x} = K_0 \left[0 - \frac{5(3yz)x}{r^7} \right]_{\vec{r}_0} = 0 \tag{27}$$

$$\frac{\partial B_y}{\partial y} = K_0 \left[\frac{3z}{r^5} - \frac{5(3yz)y}{r^7} \right]_{\vec{r}_0} = 0 \tag{28}$$

$$\frac{\partial B_y}{\partial z} = K_0 \left[\frac{3y}{r^5} - \frac{5(3yz)z}{r^7} \right]_{\vec{r}_0} = 0 \tag{29}$$

$$\frac{\partial B_z}{\partial x} = K_0 \left[\frac{-2x}{r^5} - \frac{5(2z^2 - x^2 - y^2)x}{r^7} \right]_{\vec{r}_0} = K_0 \left[\frac{-2R_e}{R_e^5} - \frac{5(-R_e^2)R_e}{R_e^7} \right] = \frac{3K_0}{R_e^4} \tag{30}$$

$$\frac{\partial B_z}{\partial y} = K_0 \left[\frac{-2y}{r^5} - \frac{5(2z^2 - x^2 - y^2)y}{r^7} \right]_{\vec{r}_0} = 0 \tag{31}$$

$$\frac{\partial B_z}{\partial z} = K_0 \left[\frac{4z}{r^5} - \frac{5(2z^2 - x^2 - y^2)z}{r^7} \right]_{\vec{r}_0} = 0 \tag{32}$$

2. Assembled Jacobian Matrix

$$J_{\vec{B}}\big|_{\vec{r}_0} = \frac{3\mu_0 M_m}{4\pi R_e^4} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \tag{33}$$

Problem 4: Closed-Form Plasma Energy Integral

The total energy E_{total} generated over descent time $[0, T_f]$ is:

$$E_{\text{total}} = A_{\text{tether}} \int_0^{T_f} j_e(t) V(t)\, dt \tag{34}$$

Given:

$$n_e(t) = n_0 \exp\left(\frac{v_z t}{H_e}\right) \tag{35}$$

$$V(t) = V_0(1 - \alpha t)^2 \tag{36}$$

Substitute into the Langmuir current equation:

$$j_e(t) = e n_0 \exp\left(\frac{v_z t}{H_e}\right) \sqrt{\frac{k_B T_e}{2\pi m_e}} \left(1 + \frac{eV_0(1-\alpha t)^2}{k_B T_e}\right)^{1/2} \tag{37}$$

For high bias voltages $eV(t) \gg k_B T_e$:

$$\left(1 + \frac{eV(t)}{k_B T_e}\right)^{1/2} \approx \sqrt{\frac{eV_0}{k_B T_e}}(1 - \alpha t) \tag{38}$$

Substitute $j_e(t)$ and $V(t)$ back into the energy integral:

$$E_{\text{total}} \approx A_{\text{tether}} e n_0 \sqrt{\frac{eV_0}{2\pi m_e}} V_0 \int_0^{T_f} \exp\left(\frac{v_z t}{H_e}\right) (1 - \alpha t)^3\, dt \tag{39}$$

Let $C_0 = A_{\text{tether}} e n_0 V_0^{3/2} \sqrt{\frac{e}{2\pi m_e}}$ and $a = \frac{v_z}{H_e}$.

Expanding $(1-\alpha t)^3 = 1 - 3\alpha t + 3\alpha^2 t^2 - \alpha^3 t^3$:

$$E_{\text{total}} = C_0 \int_0^{T_f} e^{at} \left(1 - 3\alpha t + 3\alpha^2 t^2 - \alpha^3 t^3\right) dt \tag{40}$$

Integrating by parts iteratively ($\int t^n e^{at} dt$):

$$E_{\text{total}} = C_0 \left[\frac{e^{at}}{a} \left((1 - 3\alpha t + 3\alpha^2 t^2 - \alpha^3 t^3) - \frac{1}{a}(-3\alpha + 6\alpha^2 t - 3\alpha^3 t^2) + \frac{1}{a^2}(6\alpha^2 - 6\alpha^3 t) - \frac{1}{a^3}(-6\alpha^3) \right) \right]_0^{T_f} \tag{41}$$

Evaluating at limits $t = T_f$ and $t = 0$:

$$\begin{aligned} E_{\text{total}} = C_0 \Bigg[& \frac{e^{aT_f}}{a} \left((1-\alpha T_f)^3 + \frac{3\alpha}{a}(1-\alpha T_f)^2 + \frac{6\alpha^2}{a^2}(1-\alpha T_f) + \frac{6\alpha^3}{a^3} \right) \\ & - \frac{1}{a}\left(1 + \frac{3\alpha}{a} + \frac{6\alpha^2}{a^2} + \frac{6\alpha^3}{a^3}\right) \Bigg] \end{aligned} \tag{42}$$

Problem 5: Hamiltonian Mechanics, Linearization, and Control Synthesis

(a) Hamilton's Canonical Equations

The Hamiltonian is:

$$H(r, \theta, p_r, p_\theta) = \frac{p_r^2}{2M_0} + \frac{p_\theta^2}{2M_0 r^2} - \frac{GM_p M_0}{r} \tag{43}$$

Evaluating canonical derivatives:

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_r}{M_0} \tag{44}$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{M_0 r^2} \tag{45}$$

$$\dot{p}_r = -\frac{\partial H}{\partial r} = \frac{p_\theta^2}{M_0 r^3} - \frac{GM_p M_0}{r^2} \tag{46}$$

$$\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = 0 \tag{47}$$

Since $\frac{\partial H}{\partial \theta} = 0$, θ is a cyclic coordinate, proving $p_\theta = M_0 r^2 \dot{\theta} = \text{constant}$.

(b) Linearization around Circular Orbit Equilibrium

At circular equilibrium $r = r_0$, $p_{r0} = 0$, and $\dot{p}_r = 0$:

$$\frac{p_{\theta 0}^2}{M_0 r_0^3} - \frac{GM_p M_0}{r_0^2} = 0 \implies r_0 = \frac{p_{\theta 0}^2}{GM_p M_0^2} \tag{48}$$

Let perturbations be $x = \begin{bmatrix} \Delta r \\ \Delta p_r \end{bmatrix}$. Linearizing $\begin{bmatrix} \dot{r} \\ \dot{p}_r \end{bmatrix} = \begin{bmatrix} f_1(r, p_r) \\ f_2(r, p_r) \end{bmatrix}$:

$$\frac{\partial f_1}{\partial r} = 0, \quad \frac{\partial f_1}{\partial p_r} = \frac{1}{M_0} \tag{49}$$

$$\frac{\partial f_2}{\partial r} = -\frac{3p_{\theta 0}^2}{M_0 r_0^4} + \frac{2GM_p M_0}{r_0^3} = -\frac{3GM_p M_0}{r_0^3} + \frac{2GM_p M_0}{r_0^3} = -\frac{GM_p M_0}{r_0^3} \tag{50}$$

$$\frac{\partial f_2}{\partial p_r} = 0 \tag{51}$$

Constructing system matrix A:

$$A = \begin{bmatrix} 0 & \frac{1}{M_0} \\ -\frac{GM_p M_0}{r_0^3} & 0 \end{bmatrix} \tag{52}$$

(c) Eigenvalues, Resonance, and State-Feedback Synthesis

Compute eigenvalues $\det(\lambda I - A) = 0$:

$$\det \begin{bmatrix} \lambda & -\frac{1}{M_0} \\ \frac{GM_p M_0}{r_0^3} & \lambda \end{bmatrix} = \lambda^2 + \frac{GM_p}{r_0^3} = 0 \implies \lambda_{1,2} = \pm i\sqrt{\frac{GM_p}{r_0^3}} = \pm i\omega_0 \tag{53}$$

where $\omega_0 = \sqrt{\frac{GM_p}{r_0^3}}$ is the natural orbital angular frequency.

Adding radial control input $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_c$ with feedback $u_c = -Kx = -k_1\Delta r - k_2\Delta p_r$:

$$A_{\text{cl}} = A - BK = \begin{bmatrix} 0 & \frac{1}{M_0} \\ -\omega_0^2 M_0 - k_1 & -k_2 \end{bmatrix} \tag{54}$$

Closed-loop characteristic equation:

$$\det(\lambda I - A_{\text{cl}}) = \lambda^2 + k_2\lambda + \left(\omega_0^2 + \frac{k_1}{M_0}\right) = 0 \tag{55}$$

Equating to target polynomial $P_{\text{target}}(\lambda) = (\lambda - \mu_1)(\lambda - \mu_2) = \lambda^2 - (\mu_1 + \mu_2)\lambda + \mu_1\mu_2$:

$$k_2 = -(\mu_1 + \mu_2) \tag{56}$$

$$\omega_0^2 + \frac{k_1}{M_0} = \mu_1\mu_2 \implies k_1 = M_0(\mu_1\mu_2 - \omega_0^2) \tag{57}$$

Thus, feedback gains $K = \begin{bmatrix} M_0(\mu_1\mu_2 - \omega_0^2) & -(\mu_1 + \mu_2) \end{bmatrix}$ uniquely assign stable closed-loop poles at $\{\mu_1, \mu_2\} \subset \mathbb{R}^-$.